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Velocity-Based Training With Weightlifting Derivatives: Barbell and System Velocity Comparisons

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[1]Department of Sports Medicine and Nutrition, University of Pittsburgh, Pittsburgh, Pennsylvania; [2]Department of Human Movement Sciences, Carroll University, Waukesha, Wisconsin; [3]Directorate of Psychology and Sport, University of Salford, Salford, Greater Manchester, United Kingdom; [4]New York Mets, Queens, New York; [5]Athlete Performance, Mequon, Wisconsin; [6]Department of Kinesiology and Sport Management, Texas A&M University, College Station, Texas; and [7]Strength and Power Research Group, School of Medical and Health Sciences, Edith Cowan University, Joondalup, Western Australia, Australia

Abstract

Suchomel, TJ, Kissick, CR, Techmanski, BS, Mann, JB, and Comfort, P. Velocity-based training with weightlifting derivatives: Barbell and system velocity comparisons. *J Strength Cond Res* 39(2): 135–146, 2025—The aim of this study was to examine the differences in barbell and system (i.e., subject + load) velocity during weightlifting derivatives performed across a spectrum of relative loads. 14 resistance-trained men participated in 6 testing sessions, which included 1 repetition maximum hang power clean (HPC) testing and individual jump shrug (JS), hang high pull (HHP), HPC, hang clean pull (HCP), and countermovement shrug (CMS) sessions. The order of the exercise testing sessions was randomized and required the subjects to perform either JS, HHP, HPC, HCP, or CMS repetitions while standing on a force platform with a linear position transducer attached to the barbell. The JS and HHP were performed with 20, 40, 60, 80, and 100% of their 1 repetition maximum HPC, HPC with 20, 40, 60, and 80% 1RM, and the HCP and CMS performed with 20, 40, 60, 80, 100, 120, and 140% 1RM. Mean and peak barbell and system velocities were determined across all exercises and loads using either 2×5, 2×4, or 2×7 repeated measures ANOVA depending on the number of loads performed. Significantly ($p < 0.001$) and meaningfully ($g \geq 1.49$) greater mean and peak barbell velocities existed at every exercise and load combination compared with the mean and peak system velocities produced. Barbell and system velocity are distinct characteristics that should not be substituted for one another. Owing to the characteristics of the transition phase, mean barbell and system velocity may not provide strength and conditioning practitioners with meaningful information related to load prescription.

Key Words: jump shrug, hang high pull, hang power clean, hang pull, countermovement shrug, Olympic weightlifting

Introduction

The use of weightlifting movements (snatch and clean and jerk) and their derivatives in resistance training programs is well documented (3,27). Researchers have shown that training with weightlifting movements may produce superior strength-power adaptations compared with traditional resistance training exercises (1,22), jump training (22), and kettlebell training (23). Beyond weightlifting catching variations such as the hang power clean (HPC) (30), pulling derivatives (i.e., weightlifting variations performed from various starting positions that omit the catch phase) such as the jump shrug (JS) (35), hang high pull (HHP) (34), hang clean pull (HCP) (7), and countermovement shrug (CMS) (6) may provide unique training stimuli for athletes (28,32,33). For example, researchers have shown that training with weightlifting pulling derivatives may produce similar (2) or greater (38–40) strength-power adaptations likely due to their ability to provide individuals with a greater force or velocity overload stimulus (32). In fact, greater force and velocity characteristics were produced during the JS and HHP compared with the HPC (12–14,44,45,49–51). In addition, the HCP (20) and CMS (21) may allow individuals to use loads up to at least 140% of a 1 repetition maximum (1RM) catching variation potentially leading to greater force-related adaptations. Although there appears to be a strong rationale for the implementation of weightlifting exercises and their derivatives, there is limited information on how to load pulling variations beyond the use of relative 1RM catching derivative percentages, and thus, further information on potential loading methods may be beneficial to practitioners.

A wide variety of loading methods may be used when prescribing resistance training exercises (43). One method that has gained considerable popularity has been termed velocity-based training (VBT). While there are several potential uses for VBT (54), one of its primary uses has been the implementation of VBT zones whereby resistance training loads are prescribed and adjusted based on a given range of velocity. For example, Mann (18) provided a training spectrum of different mean barbell velocity (MBV) zones during the back squat exercise that coincide with different 1RM percentages and training foci. While these zones may be effective for training prescription with traditional exercises (e.g., squats, presses, and pulls), less

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is known about peak barbell velocity (PBV) and VBT with weightlifting derivatives. Haff and colleagues (8) indicated that while PBV may allow strength and conditioning practitioners to estimate the 1RM power clean of an individual, monitoring changes in maximal strength may be challenging due to the coefficients of variation present with both stronger (7.5–7.6%) and weaker individuals (11.6–13.3%). Moreover, the authors concluded that MBV may not provide an accurate assessment, which is likely due to the highly technical nature of weightlifting derivatives and the intent to maximize barbell velocity (25). These findings are in line with LeMense et al. (17) who have challenged the accuracy of using load-velocity relationships to predict a 1RM. Owing to the lack of research on measuring barbell velocity with weightlifting exercises and their derivatives (8,47,48,52,53) and its potential use for load prescription with pulling derivatives, further research is needed if practitioners want to use VBT for load prescription with these exercises.

There are a wide variety of methods that have been used to examine the force, velocity, and power characteristics of weightlifting derivatives. For example, researchers have shown considerable differences in power output during the HPC or power clean exercise when these values are calculated using either force plate technology, linear position transducers, or a combination of both (19,26). These findings may be explained by the fact that the barbell travels independently from the body during these exercises. It should be noted that while the force applied to the barbell, and by extension the resultant barbell velocity, may be important for a competitive weightlifter when it comes to making their lifts, this concept may be less important for non-weightlifters who may be more concerned with how the entire system (athlete + load) is moving (19). However, because weightlifting pulling derivatives omit the catch phase, the differences between barbell and system velocity are unknown for these exercises. Given their potential as training stimuli, this information may aid practitioners in choosing different exercise and load combinations to improve the force and/or velocity characteristics of the individuals that they train. In this light, the purpose of this study was to compare barbell and system velocity between several weightlifting exercises performed across a spectrum of loads. Owing to the displacement of the barbell, it was hypothesized that both MBV and PBV would be significantly greater than mean system velocity (MSV) and peak system velocity (PSV) across the loading spectrum of each exercise, respectively, in line with previous research using different exercises (16,19).

Methods

Experimental Approach to the Problem

A repeated measures, cross-sectional design was used to examine the differences between mean and peak barbell and system velocity during the JS, HHP, HPC, HCP, and CMS exercises performed across a series of relative loads. Subjects participated in 6 different testing sessions completed across approximately 2 weeks (Figure 1). Each exercise was performed using percentages of the individuals' 1 repetition maximum hang power clean (1RM HPC).

Subjects

The subjects in this study included 14 resistance-trained men (age = 25.6 ± 4.7 years [range: 20–33], body mass = 88.7 ± 16.3 kg, height = 175.5 ± 8.6 cm, relative 1RM HPC = 1.28 ± 0.24 $kg \cdot kg^{-1}$, and relative 1RM back squat = 2.05 ± 0.30 $kg \cdot kg^{-1}$) that had previous experience with the HPC and its derivatives but were not competitive weightlifters. Each subject had been consistently training at least 3 times per week for the previous year and incorporated various weightlifting derivatives into their training at least once per week. The institutional review board of Carroll University approved this study (#19-015; approved April 15, 2019), and each subject provided written informed consent before participation.

An a priori sample size estimation was completed using G*Power (version 3.1.9.4) for the repeated measures design used in this study. At a statistical power of 0.90, a sample size of at least 12 subjects was needed to display a large effect size (i.e., Hedge's $g \geq 1.20$) between barbell and system velocity measurements. Although the effect sizes displayed by McBride et al. (19) when comparing peak velocity between the barbell and system were considerably large (d = 6.44–8.94), it should be noted their study examined the power clean exercise performed from the floor which would produce larger displacements compared with the exercises in this study, which in turn may affect the resulting velocities. Thus, it was expected that the effect sizes in this study would be smaller.

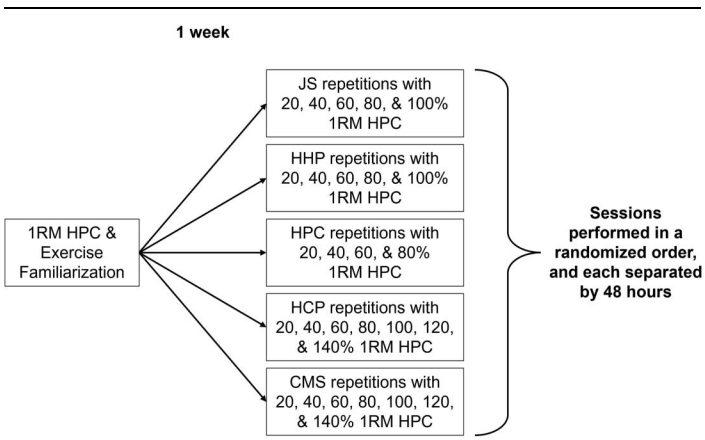

Figure 1. Testing sequence and timeline.

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Figure 2. Weightlifting derivative exercise testing session setup.

Procedures

One Repetition Maximum Hang Power Clean Testing Session. The subjects completed 1RM HPC testing during their first session using previously described procedures (44,45,49). After completing the informed consent process, the subjects were weighed before performing a standardized warm-up. In brief, the subjects completed a general warm-up (3 minutes of light-moderate cycling followed by dynamic stretching) before they performed warm-up HPC sets with 30, 50, 70, and 90% of their self-estimated 1RM. Following the final warm-up repetition, the principal investigator and research assistants adjusted the load for 1RM attempts increasing the load by at least 2.5 kg. Each subject then performed maximal 1RM HPC attempts with 3–5 minutes of rest between lifts until a failed attempt occurred; the 1RM in this case was determined as the previous load they successfully completed. A successful attempt required the subjects to complete the HPC lift without the top of their thighs dropping below parallel during the catch phase. Each attempt was visually monitored to ensure the lift was performed in an appropriate manner. The 1RM of each subject was established in 4 repetitions or fewer. Following the 1RM test, subjects were provided with a self-selected rest period before they were familiarized with the JS, HHP, HCP, and CMS exercises. Familiarization repetitions were performed with light loads (<50% 1RM HPC) using the coaching cues discussed in previous studies (6,7,30,34,35).

Exercise Testing Sessions. The first of 5 exercise testing sessions took place 1 week following the 1RM HPC session, and each session was separated by 48 hours. During the testing week, the

Table 1

Intraclass correlation coefficients and coefficients of variation percentages with 95% confidence intervals for mean and peak barbell velocities during the jump shrug, hang high pull, hang power clean, hang clean pull, and countermovement shrug.*†

	Mean barbell velocity		Peak barbell velocity	
Load (%)	**ICC**	**CV%**	**ICC**	**CV%**
Jump shrug				
20	0.84 (0.62–0.95)	7.3 (5.3–12.1)	0.95 (0.88–0.98)	3.5 (2.6–5.8)
40	0.93 (0.82–0.97)	4.1 (3.0–6.7)	0.95 (0.88–0.98)	3.1 (2.2–5.0)
60	0.91 (0.79–0.97)	3.7 (2.7–6.0)	0.95 (0.88–0.98)	2.0 (1.4–3.2)
80	0.93 (0.81–0.98)	5.1 (3.6–8.3)	0.93 (0.83–0.98)	3.6 (2.6–5.8)
100	0.97 (0.91–0.99)	3.9 (2.8–6.3)	0.90 (0.74–0.96)	3.4 (2.4–5.5)
Hang high pull				
20	0.95 (0.88–0.98)	5.9 (4.2–9.7)	0.96 (0.89–0.98)	4.8 (3.4–7.8)
40	0.97 (0.93–0.99)	3.2 (2.3–5.1)	0.93 (0.83–0.98)	3.2 (2.3–5.2)
60	0.95 (0.88–0.98)	4.7 (3.4–7.7)	0.96 (0.90–0.99)	2.9 (2.1–4.7)
80	0.97 (0.94–0.99)	3.2 (2.3–5.2)	0.98 (0.95–0.99)	2.6 (1.9–4.2)
100	0.98 (0.94–0.99)	3.2 (2.3–5.3)	0.98 (0.94–0.99)	2.0 (1.5–3.3)
Hang power clean				
20	0.88 (0.71–0.96)	7.8 (5.6–12.9)	0.93 (0.84–0.98)	5.6 (4.0–9.1)
40	0.95 (0.87–0.98)	4.4 (3.1–7.1)	0.78 (0.47–0.92)	4.2 (3.0–6.8)
60	0.97 (0.93–0.99)	3.4 (2.5–5.5)	0.90 (0.75–0.96)	3.3 (2.3–5.3)
80	0.92 (0.80–0.97)	5.2 (3.8–8.5)	0.90 (0.74–0.97)	4.1 (3.0–6.7)
Hang clean pull				
20	0.98 (0.95–0.99)	3.6 (2.6–5.9)	0.94 (0.83–0.98)	3.6 (2.6–5.9)
40	0.97 (0.91–0.99)	4.2 (3.0–6.8)	0.90 (0.74–0.96)	3.9 (2.8–6.4)
60	0.97 (0.93–0.99)	3.3 (2.4–5.4)	0.95 (0.87–0.98)	2.9 (2.1–4.8)
80	0.96 (0.90–0.99)	3.5 (2.5–5.6)	0.93 (0.83–0.98)	3.7 (2.7–6.0)
100	0.97 (0.92–0.99)	3.5 (2.5–5.6)	0.93 (0.83–0.98)	2.8 (2.0–4.5)
120	0.94 (0.86–0.98)	4.3 (3.1–7.1)	0.88 (0.69–0.96)	3.1 (2.2–5.0)
140	0.96 (0.89–0.99)	4.2 (3.0–6.9)	0.90 (0.76–0.97)	3.7 (2.7–6.0)
Countermovement shrug				
20	0.90 (0.75–0.96)	3.9 (2.8–6.4)	0.87 (0.68–0.96)	3.6 (2.6–5.8)
40	0.86 (0.67–0.95)	4.9 (3.5–8.0)	0.91 (0.77–0.97)	3.5 (2.5–5.7)
60	0.91 (0.77–0.97)	5.2 (3.8–8.6)	0.95 (0.88–0.98)	2.5 (1.8–4.0)
80	0.96 (0.91–0.99)	3.6 (2.6–5.8)	0.92 (0.79–0.97)	5.3 (3.8–8.8)
100	0.92 (0.81–0.97)	3.8 (2.7–6.2)	0.91 (0.79–0.97)	3.0 (2.2–4.9)
120	0.90 (0.76–0.97)	4.5 (3.3–7.4)	0.88 (0.69–0.96)	3.6 (2.6–5.9)
140	0.94 (0.84–0.98)	4.5 (3.2–7.5)	0.94 (0.83–0.98)	4.2 (3.0–7.0)

*ICC = intraclass correlation coefficients; CV% = coefficients of variation percentages.
†Loads are based on percentages of the 1 repetition maximum hang power clean of the subjects.

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subjects were instructed to avoid strenuous exercise that would impact their testing results at least 24 hours before their session; however, they were still allowed to lift during this study. The order of the testing sessions was randomized to prevent an order effect and required the subject to perform sets of either the JS, HHP, HPC, HCP, or CMS exercise using loads relative to their 1RM HPC. Each testing session began with the subject performing the same standardized warm-up described previously. Afterward, subjects performed a self-selected warm-up with an empty 20 kg barbell before completing 3 repetitions of each of the exercises that were being tested that session with 30 and 50% of their 1RM HPC (37,46). Following a 2-minute rest period, subjects began performing their maximal effort JS, HHP, HPC, HCP, or CMS repetitions. The JS and HHP repetitions were performed with 20, 40, 60, 80, and 100% of their 1RM HPC, while the HPC was performed with 20, 40, 60, and 80%, and the HCP and CMS used 20, 40, 60, 80, 100, 120, and 140% 1RM HPC. While the authors acknowledge that the lighter loads examined would not be prescribed with competitive weightlifters, the spectrum of loads was chosen based on the loading potential of each exercise and to determine how the performance of each exercise changed with non-weightlifters. The loads were performed in a progressive manner to increase ecological validity, and 3 repetitions were performed with each load. One minute of rest was provided between each repetition while 2 minutes of rest separated each load. To maintain the safety of the subjects, 80% 1RM was chosen as the highest load chosen for the HPC. Moreover, it was not practical to require a subject to perform multiple repetitions of their 1RM load during testing. However, based on previous research, the JS and HHP exercises were performed with loads up to 100% 1RM (47) while the HCP and CMS used loads up to 140% 1RM (20,21). To prevent grip from being a limiting factor, lifting straps were used during all repetitions performed at 100, 120, and 140% 1RM. The testing repetitions were performed based on previous descriptions (6,7,30,34,35). Apart from the CMS, subjects started each repetition by standing in the mid-thigh (power) position before receiving the countdown of "3, 2, 1, Go!" Upon hearing the "Go!" command, subjects performed a hip hinge movement and lowered the barbell down their thighs until the barbell was just above their patellae and without pausing, transitioned back to the mid-thigh position by flexing their knees, elevating their chest, and moving the barbell back up their thighs. Upon reaching the mid-thigh position, the subjects rapidly extended their hip, knee, and ankle (plantarflexion) joints to perform the second pull

Table 2

Intraclass correlation coefficients and coefficients of variation percentages with 95% confidence intervals for mean and peak system velocities during the jump shrug, hang high pull, hang power clean, hang clean pull, and countermovement shrug.*†

	Mean system velocity		Peak system velocity	
Load (%)	ICC	CV%	ICC	CV%
Jump shrug				
20	0.87 (0.69–0.96)	11.7 (8.4–19.6)	0.96 (0.91–0.99)	2.3 (1.7–3.8)
40	0.93 (0.82–0.97)	5.2 (3.7–8.5)	0.96 (0.90–0.99)	2.5 (1.8–4.0)
60	0.91 (0.77–0.97)	4.4 (3.1–7.1)	0.95 (0.88–0.98)	2.4 (1.7–3.8)
80	0.94 (0.86–0.98)	4.9 (3.5–8.0)	0.97 (0.92–0.99)	3.0 (2.2–4.9)
100	0.95 (0.88–0.98)	4.3 (3.1–7.1)	0.94 (0.86–0.98)	2.9 (2.1–4.8)
Hang high pull				
20	0.99 (0.98–0.99)	5.4 (3.9–8.8)	0.92 (0.80–0.97)	5.4 (3.9–8.8)
40	0.99 (0.97–0.99)	3.6 (2.6–5.8)	0.97 (0.92–0.99)	2.9 (2.1–4.7)
60	0.96 (0.91–0.99)	6.0 (4.3–9.9)	0.92 (0.81–0.97)	3.9 (2.8–6.3)
80	0.96 (0.91–0.99)	5.2 (3.7–8.4)	0.97 (0.92–0.99)	3.0 (2.2–4.9)
100	0.96 (0.91–0.99)	5.7 (4.1–9.3)	0.96 (0.89–0.98)	3.0 (2.2–4.9)
Hang power clean				
20	0.95 (0.88–0.98)	8.3 (5.9–13.7)	0.95 (0.88–0.98)	7.8 (5.6–12.9)
40	0.98 (0.96–0.99)	4.1 (3.0–6.7)	0.90 (0.76–0.97)	6.2 (4.5–10.2)
60	0.98 (0.94–0.99)	4.0 (2.9–6.5)	0.94 (0.84–0.98)	4.0 (2.9–6.5)
80	0.97 (0.93–0.99)	4.7 (3.4–7.7)	0.94 (0.85–0.98)	4.6 (3.3–7.5)
Hang clean pull				
20	0.99 (0.96–0.99)	5.2 (3.7–8.4)	0.79 (0.47–0.93)	6.6 (4.7–10.8)
40	0.99 (0.98–0.99)	3.8 (2.7–6.1)	0.83 (0.59–0.94)	4.8 (3.5–7.9)
60	0.98 (0.96–0.99)	3.4 (2.5–5.6)	0.92 (0.79–0.97)	4.1 (2.9–6.6)
80	0.98 (0.94–0.99)	3.6 (2.6–5.8)	0.95 (0.88–0.98)	3.9 (2.8–6.4)
100	0.95 (0.87–0.98)	5.9 (4.3–9.7)	0.91 (0.78–0.97)	3.8 (2.7–6.1)
120	0.96 (0.91–0.99)	4.4 (3.2–7.2)	0.96 (0.90–0.99)	3.6 (2.6–5.8)
140	0.97 (0.93–0.99)	3.6 (2.6–5.8)	0.92 (0.81–0.97)	5.4 (3.9–8.9)
Countermovement shrug				
20	0.89 (0.74–0.96)	5.9 (4.2–9.6)	0.77 (0.43–0.92)	5.4 (3.9–8.8)
40	0.91 (0.78–0.97)	4.6 (3.3–7.4)	0.87 (0.68–0.95)	4.2 (3.0–6.9)
60	0.96 (0.90–0.99)	3.5 (2.5–5.7)	0.95 (0.89–0.98)	2.1 (1.5–3.5)
80	0.97 (0.92–0.99)	4.6 (3.3–7.4)	0.93 (0.84–0.98)	5.0 (3.6–8.2)
100	0.96 (0.90–0.99)	4.0 (2.9–6.5)	0.94 (0.86–0.98)	3.2 (2.3–5.3)
120	0.93 (0.83–0.98)	4.6 (3.3–7.4)	0.90 (0.74–0.96)	4.2 (3.0–6.9)
140	0.97 (0.92–0.99)	4.2 (3.1–6.9)	0.96 (0.89–0.98)	4.1 (2.9–6.7)

*ICC = intraclass correlation coefficients; CV% = coefficients of variation percentages.
†Loads are based on percentages of the 1 repetition maximum hang power clean of the subjects.

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movement. While similar, the primary differences between the exercises were that the subjects jumped as high as possible, elevated the barbell to chest height by flexing the elbows, dropped under and racked the barbell across their shoulders in a front squat position, or shrugged their shoulders during the JS (35), HHP (34), HPC (30), and HCP (7) exercises, respectively. The CMS required the subjects to stand in an upright position and upon receiving the same countdown, drop to the mid-thigh position while maintaining a vertical torso, rapidly extend their hip, knee, and ankle joints, and shrug their shoulders (6). The subjects received strong verbal encouragement throughout each testing session to ensure maximal effort.

Data Analyses. Every weightlifting derivative repetition was performed with a linear position transducer (GymAware Powertool, Kinetic Performance Technology, Braddon, Australia) attached to the barbell and while standing on a force platform (Model 6090-06, Bertec Corporation, Columbus, OH) (Figure 2). While the force platform sampled at 1,000 Hz, the linear position transducer was connected through Bluetooth to a tablet (iPad 2, Apple Inc., Cupertino, CA) that used the GymAware application (version 3.1) and used a variable sampling rate with level crossing detection (47). Using its cable and spoon design, the linear position transducer measured the vertical displacement (while accounting for horizontal displacement) of each repetition and calculated both MBV and PBV by dividing the displacement by the movement time. The system velocity of each repetition was calculated using a forward dynamics approach. Using this method, raw force-time data from the force platform were integrated to calculate acceleration-time and velocity-time data. The onset of each exercise was determined as the point at which force dropped below 5 times the *SD* of the standing system mass (24). In addition, the start of the braking phase was determined as the point at which force returned to system mass. Finally, the propulsive phase was defined as the point where velocity became positive following the unweighting phase and braking phase of each repetition and ended when force dropped below system mass. Similar to barbell velocity, MSV and PSV during the propulsive phase of each repetition were then determined as the average and highest values during this period, respectively.

Statistical Analyses

The normal distribution of the dependent variables was assessed using the Shapiro-Wilks test. In addition, the relative and absolute reliability of the data was examined with 2-way

Table 3

Mean and peak barbell and system velocities produced during the jump shrug, hang high pull, hang power clean, hang clean pull, and countermovement shrug.*

	Mean velocity ($m \cdot s^{-1}$)			Peak velocity ($m \cdot s^{-1}$)		
Load	Barbell	System	*g*	Barbell	System	*g*
Jump shrug						
20	1.49 ± 0.15	1.20 ± 0.18	1.76 (0.89–2.63)	2.93 ± 0.24	2.38 ± 0.17	2.56 (1.56–3.56)
40	1.41 ± 0.11	1.11 ± 0.13	2.42 (1.45–3.40)	2.56 ± 0.21	2.17 ± 0.15	2.09 (1.17–3.01)
60	1.26 ± 0.10	1.02 ± 0.08	2.55 (1.55–3.55)	2.22 ± 0.14	1.91 ± 0.12	2.30 (1.34–3.25)
80	1.09 ± 0.12	0.87 ± 0.09	2.03 (1.12–2.94)	1.88 ± 0.16	1.64 ± 0.15	1.49 (0.65–2.33)
100	1.04 ± 0.12	0.85 ± 0.09	1.79 (0.91–2.67)	1.77 ± 0.10	1.55 ± 0.10	2.08 (1.16–3.00)
Hang high pull						
20	1.79 ± 0.25	0.75 ± 0.20	4.45 (3.07–5.83)	2.92 ± 0.33	1.58 ± 0.17	4.92 (3.43–6.40)
40	1.60 ± 0.19	0.80 ± 0.16	4.39 (3.02–5.75)	2.52 ± 0.18	1.68 ± 0.15	4.96 (3.46–6.45)
60	1.37 ± 0.16	0.77 ± 0.13	4.02 (2.73–5.31)	2.25 ± 0.17	1.64 ± 0.13	3.83 (2.59–5.08)
80	1.16 ± 0.13	0.70 ± 0.10	3.72 (2.50–4.95)	1.97 ± 0.19	1.51 ± 0.15	2.62 (1.61–3.63)
100	1.05 ± 0.12	0.68 ± 0.10	3.21 (2.09–4.33)	1.82 ± 0.14	1.46 ± 0.12	2.79 (1.75–3.83)
Hang power clean						
20	1.46 ± 0.19	0.62 ± 0.13	4.99 (3.49–6.49)	2.45 ± 0.29	1.23 ± 0.22	4.58 (3.17–5.98)
40	1.38 ± 0.16	0.70 ± 0.13	4.56 (3.16–5.97)	2.28 ± 0.15	1.39 ± 0.16	5.56 (3.92–7.19)
60	1.29 ± 0.14	0.72 ± 0.11	4.35 (2.99–5.71)	2.17 ± 0.12	1.49 ± 0.13	5.38 (3.79–6.98)
80	1.15 ± 0.10	0.69 ± 0.10	4.51 (3.12–5.91)	1.95 ± 0.12	1.44 ± 0.12	4.11 (2.81–5.42)
Hang clean pull						
20	1.33 ± 0.22	0.66 ± 0.16	3.36 (2.21–4.51)	2.34 ± 0.23	1.39 ± 0.11	5.18 (3.64–6.73)
40	1.25 ± 0.16	0.69 ± 0.14	3.53 (2.35–4.72)	2.18 ± 0.15	1.44 ± 0.09	5.78 (4.09–7.46)
60	1.15 ± 0.14	0.69 ± 0.12	3.49 (2.32–4.67)	2.00 ± 0.15	1.45 ± 0.11	4.14 (2.82–5.45)
80	1.05 ± 0.11	0.66 ± 0.09	3.57 (2.38–4.77)	1.80 ± 0.14	1.37 ± 0.14	2.95 (1.88–4.02)
100	1.00 ± 0.12	0.65 ± 0.09	3.20 (2.08–4.32)	1.74 ± 0.10	1.37 ± 0.11	3.56 (2.37–4.75)
120	0.91 ± 0.10	0.61 ± 0.08	3.17 (2.05–4.28)	1.59 ± 0.09	1.27 ± 0.12	2.84 (1.79–3.88)
140	0.81 ± 0.09	0.56 ± 0.07	2.94 (1.87–4.01)	1.41 ± 0.09	1.16 ± 0.12	2.26 (1.31–3.21)
Countermovement shrug						
20	1.37 ± 0.10	0.86 ± 0.07	5.56 (3.93–7.19)	2.15 ± 0.13	1.41 ± 0.08	6.73 (4.82–8.64)
40	1.33 ± 0.10	0.90 ± 0.08	4.52 (3.13–5.92)	2.06 ± 0.15	1.47 ± 0.10	4.52 (3.12–5.92)
60	1.20 ± 0.11	0.85 ± 0.07	3.65 (2.44–4.87)	1.86 ± 0.13	1.41 ± 0.08	4.03 (2.74–5.32)
80	1.04 ± 0.10	0.76 ± 0.10	2.72 (1.69–3.75)	1.63 ± 0.14	1.29 ± 0.12	2.45 (1.47–3.43)
100	1.05 ± 0.09	0.78 ± 0.08	3.08 (1.99–4.18)	1.63 ± 0.10	1.33 ± 0.09	2.97 (1.90–4.05)
120	0.96 ± 0.08	0.72 ± 0.07	3.06 (1.97–4.15)	1.51 ± 0.08	1.24 ± 0.08	3.09 (1.99–4.19)
140	0.85 ± 0.10	0.63 ± 0.09	2.25 (1.31–3.20)	1.36 ± 0.11	1.12 ± 0.11	2.11 (1.19–3.04)

*Loads are based on percentages of the 1 repetition maximum hang power clean of the subjects; *g* = Hedge's *g* effect sizes (95% lower-bound and upper-bound confidence intervals).

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mixed intraclass correlation coefficients and typical error expressed as coefficients of variation, respectively. Poor, moderate, good, and excellent intraclass correlation coefficient magnitudes coincided with values < 0.50, 0.51–0.74, 0.75–0.90, and >0.90, respectively (15) while coefficients of variation that were <10% were considered acceptable (4). A series of mixed (type × load) repeated measures ANOVA were used to examine the potential differences between mean and peak barbell and system velocities across the examined loading spectrum of each weightlifting derivative performed in this study. Specifically, 2 × 5, 2 × 4, and 2 × 7 repeated measures ANOVA with Bonferroni *post hoc* tests were used to examine the differences between barbell and system velocities during the JS and HHP, HPC, and HCP and CMS exercises, respectively. If the assumption of sphericity was violated, Greenhouse-Geisser adjusted values were reported. A criterion alpha value of $p \leq 0.05$ indicated that a statistically significant difference was present. The magnitude of the difference between the mean and *SD* of each dependent variable was determined using Hedge's *g* effect sizes. These values were interpreted as trivial, small, moderate, large, very large, and nearly perfect when effect size values were 0.00–0.19, 0.20–0.59, 0.60–1.19, 1.20–1.99, 2.00–3.99, and ≥4.00 (9). The statistical analyses were all completed using SPSS 28 (IBM, Chicago, IL).

Results

Each of the examined variables was normally distributed and the reliability results for barbell and system velocities are displayed in Tables 1 and 2, respectively. Significant type × load interaction effects existed for all mean and peak velocity variables for the JS ($p < 0.05$), HHP ($p < 0.001$), HPC ($p < 0.001$), HCP ($p < 0.001$), and CMS ($p < 0.001$). *Post hoc* analyses revealed that every MBV and PBV was significantly greater ($p < 0.001$) with large to very large differences ($g \geq 1.49$) than its MSV and PSV counterparts with every exercise and load combination. Descriptive statistics and effect size comparisons are displayed in Table 3 and Figures 3–7.

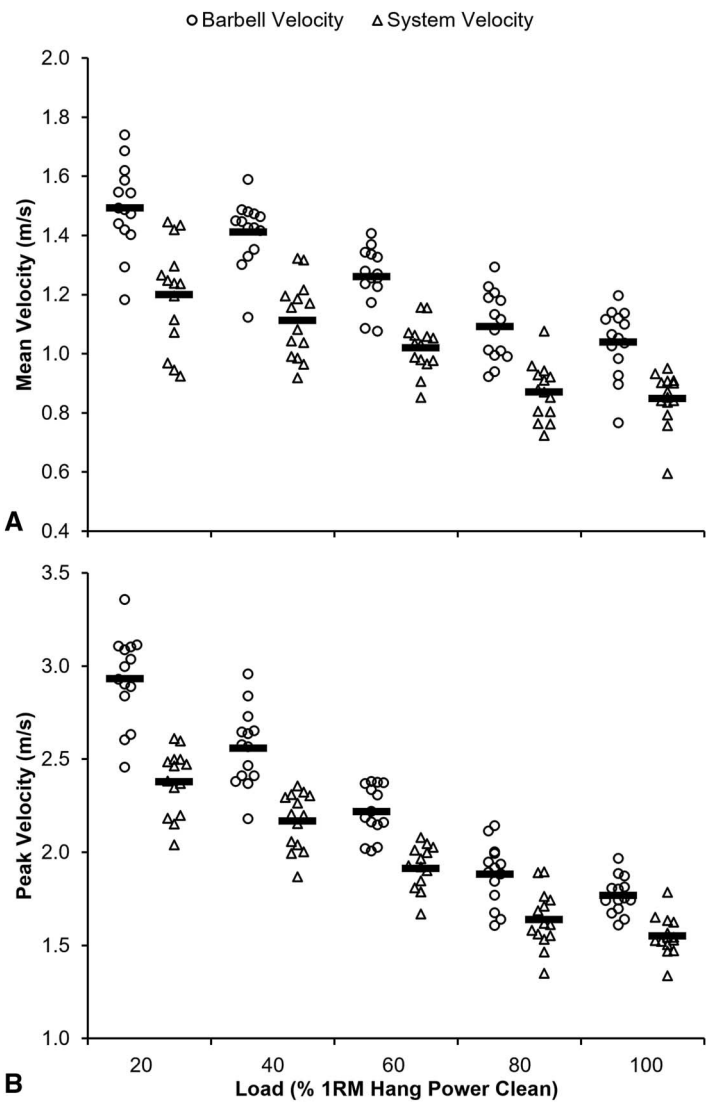

Figure 3. Group mean (black bars) and individual (open shapes) jump shrug mean (A) and peak (B) barbell and system velocities produced using relative percentages of the subjects' 1 repetition maximum hang power clean.

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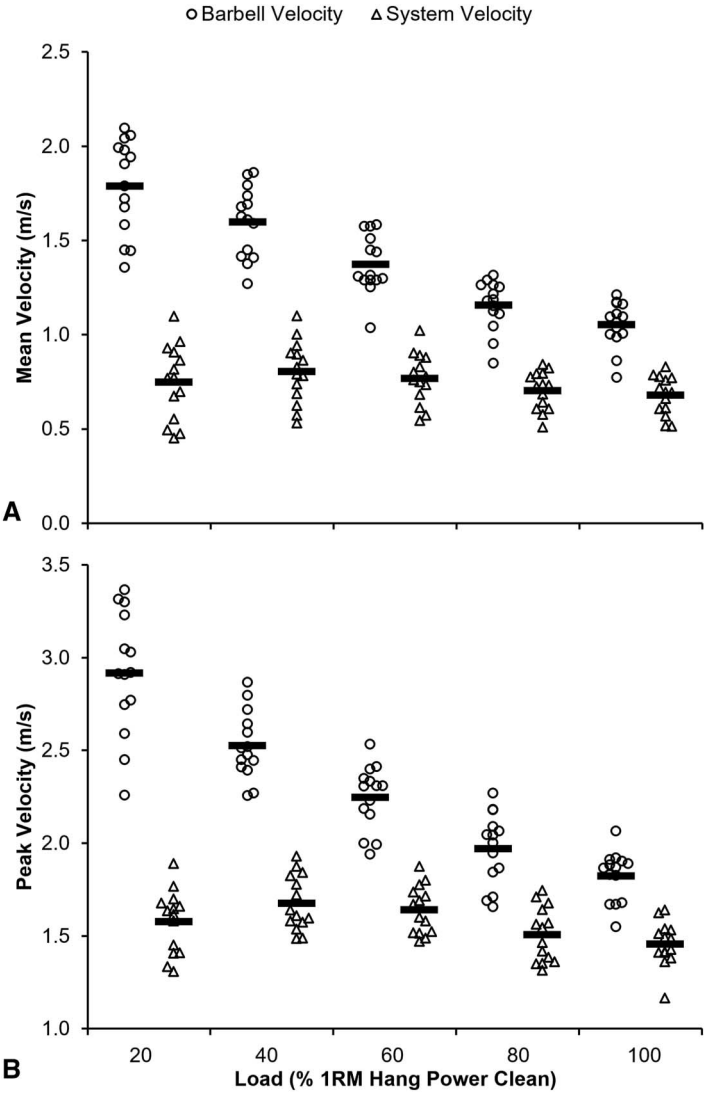

Figure 4. Group mean (black bars) and individual (open shapes) hang high pull mean (A) and peak (B) barbell and system velocities produced using relative percentages of the subjects' 1 repetition maximum hang power clean.

Discussion

The aim of this study was to compare MBV and PBV with MSV and PSV during weightlifting derivatives performed across a spectrum of loads. In line with our hypothesis, large to very large significant differences existed between barbell and system velocity within every exercise. More specifically, MBV and PBV were greater than MSV and PSV during each exercise at every load.

The large differences displayed between barbell and system velocities should not be overly surprising given that the barbell travels independently from the body during weightlifting movements. The current findings support previous work by McBride and colleagues (19) who showed that the load that maximizes peak power may be specific to the barbell, system, or joint during power cleans, squats, and jump squats. Based on their results, the previous authors concluded that the type of power (i.e., barbell, system, etc.) may be specific to the type of athlete a strength and conditioning practitioner is working with. For example, a weightlifter may need to be more focused on barbell velocity and power, whereas a non-weightlifter who uses weightlifting derivatives in training may need to focus more on what is being produced by the system. Thus, although VBT devices may provide reliable and valid barbell velocity data during weightlifting derivatives (47), it is important that practitioners understand the differences between barbell and system velocity as these variables do not provide the same information about the training stimulus an athlete receives or the quality of the movement pattern.

A primary reason for the differences between barbell and system velocities is the movement patterns that are performed during weightlifting derivatives. For example, the JS (35), HHP (34), HPC (36), and HCP (7) exercises all required the subjects to perform a hip hinge movement by lowering the barbell down their thighs before transitioning back to the mid-thigh position before the second pull phase (i.e., triple extension). Although a hip hinge may be performed quickly, the repositioning of an individual's hips and knees during the transition phase impacts the force production characteristics

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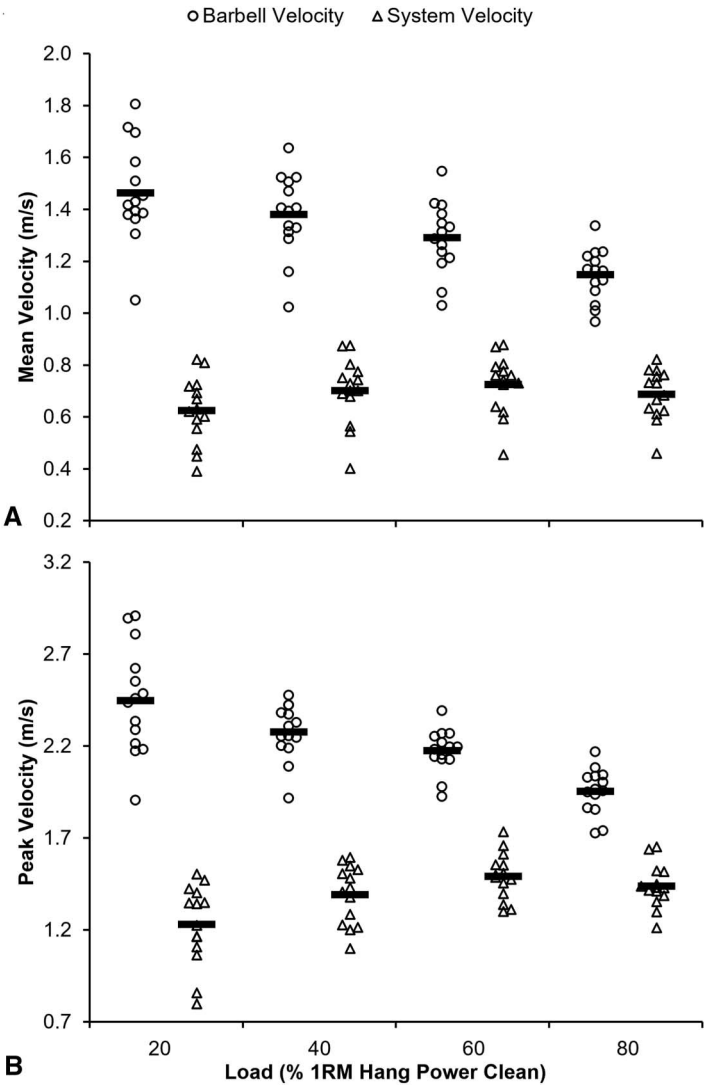

Figure 5. Group mean (black bars) and individual (open shapes) hang power clean (A) and peak (B) barbell and system velocities produced using relative percentages of the subjects' 1 repetition maximum hang power clean.

and by extension, the velocity of the movement. Previous researchers examined the force-time curve between different weightlifting derivatives and concluded that the force production characteristics were similar through the first 80–85% of the movement when using the same relative loads (12,44). However, each of the force-time curves shown in the studies displayed a "flattening" of force which coincided with the transition phase where the barbell is moving from the patellae to the mid-thigh position. From the visuals in the previous studies, it can be concluded that there is a maintenance of the system mass (i.e., subject + load) during the repositioning of the hips and knees, which also leads to minimal changes in system acceleration and thereby velocity. Collectively, this may lead to reduced MSV and PSV magnitudes. In fact, apart from the JS performed at 20, 40, and 60% 1RM HPC, all MSV were below 1.00 $m \cdot s^{-1}$. In contrast to the lower system velocities, MBV and PBV values were higher due to a constant increase in displacement starting at the lowest point of the hip hinge (i.e., above the patellae). For example, despite repositioning their hips and knees during the transition phase, the subjects continued to elevate the barbell from its initial low point throughout the movement, thus limiting reductions in velocity.

Another interesting finding of this study is centered around the similarities of the PBV of the JS and HHP across the examined loads. While both exercises may be classified as velocity-dominant weightlifting pulling derivatives (32,33), the method in which the PBV is achieved may be unique to each exercise. For example, the JS is a ballistic exercise in which the goal is to jump as high as possible (35), whereas the HHP is a semi-ballistic exercise that requires an individual to elevate the barbell to chest height (34). Thus, the ballistic nature of the JS may produce unique force production characteristics that lead to large magnitudes of system velocity (41,42,44,49). By contrast, the PBV produced during the HHP may be explained by large barbell displacements due to the requirement to elevate the barbell to chest height and additional pulling of the arms. These unique characteristics may explain the smaller

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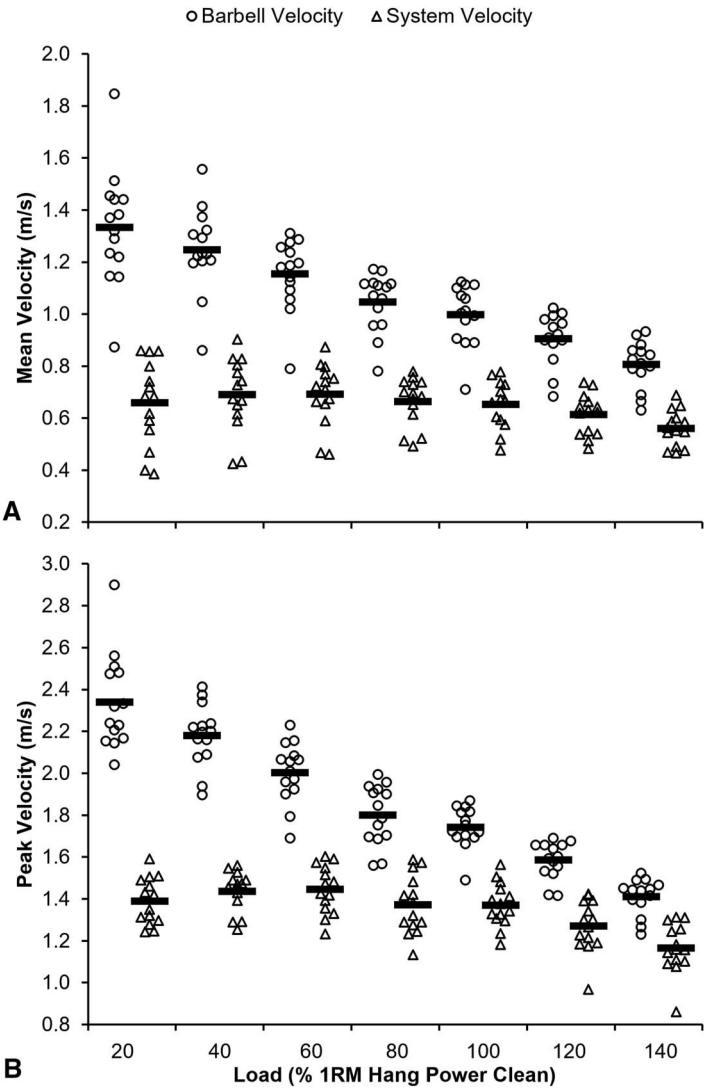

Figure 6. Group mean (black bars) and individual (open shapes) hang clean pull (A) and peak (B) barbell and system velocities produced using relative percentages of the subjects' 1 repetition maximum hang power clean.

(albeit large in magnitude) differences between PBV and PSV during the JS compared with the HHP. Furthermore, these findings should serve as a reminder that the JS may serve as a superior velocity training stimulus for athletes compared with the HHP. It should be noted however, that the differences in PSV between the JS and HHP became considerably smaller as the external load increased (20% 1RM: $g = 4.57$, 100% 1RM: $g = 0.79$), which is in line with the previous findings (49). However, given that there may be a gray area of programming weightlifting derivatives where numerous exercise and load combinations can be used to train comparable force-velocity characteristics (3,28,32), strength and conditioning practitioners may consider using various exercise and load combinations to achieve the desired training goals.

Although the subjects were encouraged to perform each exercise repetition with maximal effort, MSV and PSV were not maximized at the lightest loads for the HHP, HPC, HCP, and CMS. This is in contrast with the JS and the MBV and PBV trends of each exercise. The findings for the HHP, HPC, HCP, and CMS MSV and PSV data may be explained by the very light, submaximal loads that were being used (i.e., 20 and 40% 1RM HPC) and the movement characteristics of each exercise. Regarding the load, researchers have shown that maximal power output during the HPC, HCP, and CMS may occur at loads ranging from 60–80% (10,11,30), 100–120% (20), and 100–120% (21) 1RM HPC, respectively. While strength and conditioning practitioners may be inclined to prescribe loads within these ranges based on the previous studies, performing the exercises with lighter loads may result in individuals using submaximal effort and potentially decelerating at the end of the triple extension movement. This can especially be seen during weightlifting catching derivatives as the use of submaximal loads may permit individuals to use a reduced effort to complete the lift, which may negatively impact the training stimulus of the exercise (5). Moreover, since the HCP (20) and CMS (21) may allow athletes to use loads upward to 140% of their 1RM HPC, performing the exercises with 20% 1RM may be challenging as individuals may feel that they can simply use

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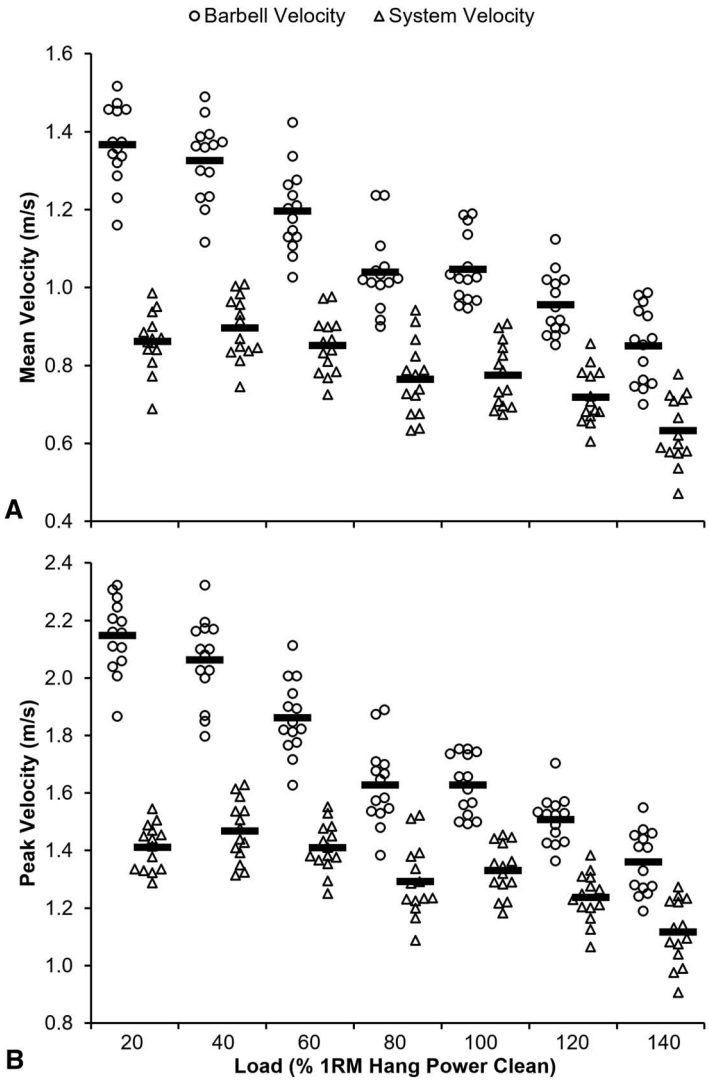

Figure 7. Group mean (black bars) and individual (open shapes) countermovement shrug (A) and peak (B) barbell and system velocities produced using relative percentages of the subjects' 1 repetition maximum hang power clean.

their arms to pull the barbell upward rather than apply the force to the ground with their legs. In addition, subjects may have had to decelerate their center of mass to prevent themselves from jumping with such a light load. The findings from Meechan et al. (20,21) support these conclusions as they showed the highest system velocities at 40% 1RM for both the HCP and CMS.

There are several limitations to this study that should be addressed. First, it should be acknowledged that some of the loads examined within each exercise may not be prescribed in a training setting. For example, loads of 20 and 40% 1RM may not be prescribed during the HPC as loads this light may limit the effort put forth during the exercise. In other words, individuals may only put forth the minimum effort necessary to elevate and catch the barbell instead of performing each repetition with maximum effort. However, it should be noted that researchers showed that maximal power output may be produced at similar lighter loads during the JS (13,29) and HHP (31), and thus, comparisons with the HPC at these loads may provide practitioners with more information about exercise selection across the loading spectrum. The same can be said for comparing the JS and HHP with the HCP and CMS at 100% 1RM HPC. Another potential limitation of this study may be the performance of individual repetitions at each load instead of multiple repetition sets. Because resistance training exercises are often prescribed with more than 1 repetition, additional research that investigates how barbell and system velocity changes across single and multiple sets of different weightlifting derivatives is needed. Finally, the subjects in this study were resistance-trained men, which may make generalizing the current results to other populations challenging. Given the impact that subject height and limb length may have on barbell velocity, researchers are encouraged to continue investigating these exercises to provide additional data that may help load prescriptions.

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Practical Applications

Barbell and system velocity are distinctly unique characteristics that should not be substituted for one another. The transition phase of the weightlifting catching and pulling derivatives may prevent the effective use of mean barbell and system velocity to be used for exercise load prescription. Owing to the ballistic/semi-ballistic intent of weightlifting derivatives, it is recommended that strength and conditioning practitioners use the peak velocity characteristics of the exercises to characterize each exercise. Barbell velocity may be an important focus for weightlifters; however, system velocity may be more important to non-weightlifters who use weightlifting catching and pulling derivatives in their resistance training programs. The results of this study may serve to guide the strength and conditioning practitioners as to which weightlifting derivative may be more effective at developing a specific characteristic. For instance, those who require greater development of the velocity end of the spectrum may benefit from utilizing the JS, whereas someone who needs to develop the force end of the spectrum may do best with the CMS.

Acknowledgments

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